

JULIAN ESTEBAN RINCÓN RODRÍGUEZ

Computer Science and Artificial Intelligence | Data Engineering · Machine Learning · Backend

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PROFESSIONAL SUMMARY

Final-year (9th semester) Computer Science and Artificial Intelligence student focused on data engineering, machine learning, and backend development. Builds end-to-end data pipelines on AWS, production predictive models, and AI systems on top of LLM APIs, with experience delivering software to paying clients and working in teams of three and four. Seeking a six-month data or software engineering internship for January – June 2027.

EDUCATION

Universidad Sergio Arboleda — Bogotá, Colombia | 2022 – Present

B.Sc. Computer Science and Artificial Intelligence | 9th of 9 semesters | GPA 4.04/5.0, with the last three semesters above 4.5/5.0.

Relevant coursework: Big Data and Data Engineering, Machine Learning, Neural Networks, Natural Language Processing, Modeling and Simulation.

PROFESSIONAL EXPERIENCE

Full-Stack and AI Developer (Freelance) — **Zafra Asesores Tributarios** | Girardot, Colombia | Jun. 2026

- Built a complete sales CRM for a tax advisory firm: FastAPI/SQLAlchemy backend with a full relational model of clients, prospects, campaigns, and interactions, plus a React/Vite frontend for prospect pipeline management.
- Integrated an AI layer using Groq (Llama 3.3 70B) for interaction summaries, email drafting, and lead scoring, along with a qualification bot on the WhatsApp Cloud API.
- Delivered and demoed the system to the client with technical documentation and a defined remaining scope (authentication and continuous deployment) ahead of production rollout.

Full-Stack Developer — **ReushiGo** | Bogotá, Colombia | May 2026

Full-stack ordering platform for a family coffee distribution business: React/Vite admin panel, Express/Prisma backend, Telegram sales bot · reushi-go.vercel.app

- Developed the admin panel (React/Vite/Tailwind) and backend (Express/Prisma), with admin-key authentication and rate limiting.
- Implemented a Telegram sales bot with interactive menus and natural-language ordering over a live 19-product catalog.
- Set up CI on GitHub Actions (lint, build, npm audit, tests) and documented a security review of the project.

SELECTED PROJECTS

NEXUS — Autonomous Personal AI System | 2025 – Present

Python · GCP · Ollama/Groq/Cerebras/NVIDIA NIM/Gemini · ONNX · ChromaDB · MCP · Tailscale

- Designed a five-tier LLM router with chained failover and 38 auto-discovered skills, running daily in production since 2025.
- Deployed a hybrid cloud/local architecture on a free-tier GCP VM (Tailscale tunnel, service account impersonation) with core services available 24/7 and automatic cost cutoff.
- Exposed system capabilities as tools for external AI agents through an MCP server, with ChromaDB vector memory and mandatory confirmation for every irreversible action.

SAVI v2 — Autonomous Real Estate Valuation System | 2026 | Team of 4

Python · XGBoost · PyTorch · Q-Learning · Double DQN · scikit-learn

- Trained an XGBoost model with a clustering feature that reached $R^2 = 0.9609$ across 20,203 properties from 2006 to 2024.
- Built a full reinforcement learning pipeline (Value Iteration, tabular Q-Learning over 8,000 episodes, and Double DQN) and a three-agent consensus policy that approves, reviews, or rejects valuations based on expected financial risk.
- Documented results in an IEEE-format paper and an interactive demo deployed on GitHub Pages.

ShopStream — AWS Big Data Pipeline | May – Jul. 2026

PySpark · AWS Lambda · EMR · Glue · RDS PostgreSQL · Flask · Zappa

- Built an end-to-end pipeline for 2.5M e-commerce events: partitioned S3 ingestion, Lambda validation (CloudWatch metrics and quarantine), EMR/PySpark analytics covering six KPIs and anomaly detection, Glue ETL, and loading into an RDS PostgreSQL warehouse.
- Reached 86% test coverage with pytest, automated CI/CD with GitHub Actions, and hardened RDS security groups for controlled access.

AWS Big Data Platform (Chinook / NYC Taxi) | Mar. – May 2026 | Team of 3

EC2 · RDS · Athena · S3 · Glue · Terraform · FastAPI · React

- Processed the NYC Yellow Taxi dataset (94M rows, 12 months) on EMR with PySpark, resolving schema inconsistencies across Parquet files via explicit StructType and unionByName normalization.
- Deployed a full-stack application (React/Vite and FastAPI on two EC2 instances, PostgreSQL on RDS) and a Hive-partitioned external table in Athena over S3.

Ayuda Terremoto Colombia — Distributed Humanitarian Coordination | Aug. 2026 – Present

FastAPI · PostgreSQL · React · Vite · IndexedDB · Render

- Developed an offline-first humanitarian coordination platform following the magnitude 7.4 earthquake in Chocó: FastAPI/PostgreSQL backend as the central node and a React/IndexedDB local node with a per-action retry sync queue.

- Implemented fuzzy-matching report deduplication, automatic activation via USGS polling, and HXL-format export; covered the backend with 80+ pytest tests.

Additional projects: Victor (Jul. – Aug. 2026), a voice race engineer for Linux sim racing (C shared-memory reader, failover across 4 LLM providers, 143 tests); an NLP API with spaCy and FastAPI on AWS Lambda and EC2 (Aug. – Sep. 2026); a custom programming language built with ANTLR4; RL and CNN labs (Q-Learning, REINFORCE, DQN, CIFAR-10).

TECHNICAL SKILLS

Languages: Python (intermediate-advanced), SQL, C, Bash, JavaScript (basic)
Machine Learning and AI: scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, spaCy, Whisper, Ollama
Big Data and Cloud: AWS (EC2, S3, RDS, Athena, Glue, EMR, Lambda), PySpark, Terraform, Docker (basic)
Data and ETL: Pandas, NumPy, Matplotlib, Power BI, Jupyter Notebook, openpyxl
Backend and Web: FastAPI, Flask, Express, React, SQLAlchemy, Prisma
Tooling: Git, GitHub Actions, pytest, ChromaDB, LLM APIs (Groq, Cerebras, NVIDIA NIM, Gemini)

ACHIEVEMENTS

- 1st Place — Wisibilízas International Competition, Universitat Pompeu Fabra | 2019**
- Led the winning web application as technical product manager and frontend/UX lead, representing Ecuador in the Senior category at age 17 against teams from Latin America and Europe.
- 1st Place — USABOT Robotics Challenge | November 2025**
- Directed the CAD design and sensor architecture of a smart automated irrigation system and delivered the commercial pitch to the judges, winning against university teams.

CERTIFICATIONS

AWS Academy Graduate — Data Engineering | Amazon Web Services Academy | May 2026 (40 hours)
Database Programming with SQL | Oracle Academy | December 2024
Database Design | Oracle Academy | August 2024

LANGUAGES

Spanish: native. English: certified B2 overall (iTEP Academic-Plus, November 2025; Listening C1, Reading and Speaking B2).